# SPATIAL AI MODEL & MACHINE LEARNING

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# 01

# Pertinent Information

# Study Area: Temecula, San Diego County, CA


## 01 BACKGROUND

Temecula is a city that experienced an exponential growth in population along with a rocketing housing market.

## 02 DATA

Vector file- acquired from OpenStreetMap (OSM) Building data. OSMBuildings.org
Raster file- acquired from Planetary Computer. planetarycomputer.microsoft.com

## 03 SOFTWARE

GeoAI- A powerful Python package for integrating artificial intelligence with geospatial data analysis and visualization.
Jupyter- Notebook style interface for programming.
Colaboratory- A hosted Jupyter Notebook service that requires no setup to use and provides free of charge access to computing resources, including GPUs and TPUs. Colab is especially well suited to machine learning, data science, and education.

# 02

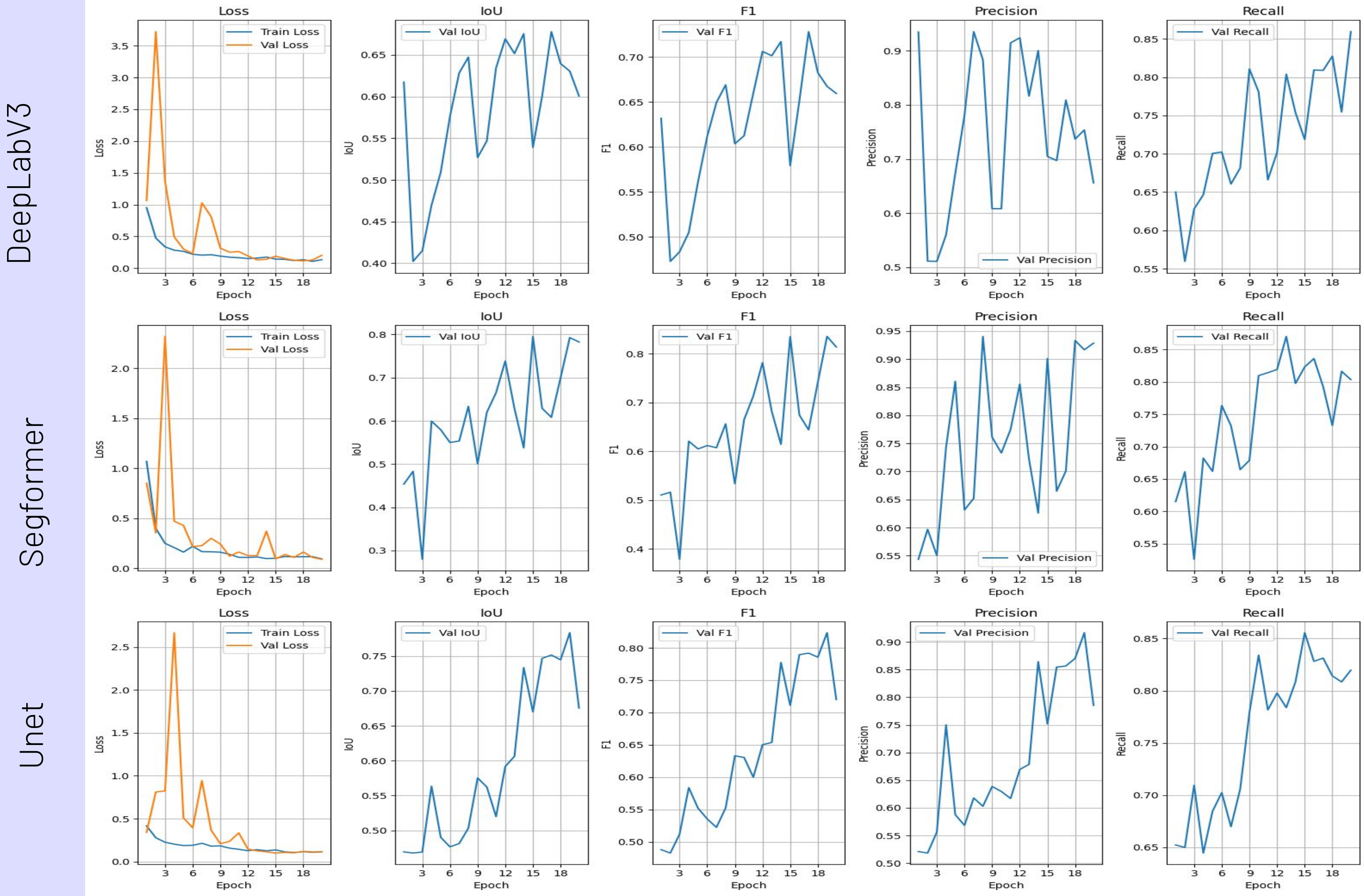
# Training

# MODEL TRAINING

- Create Training Data - Dataset is split into image chips.
  - Results: 48 total, 34 with features.
- Segmentation Models- all models use encoder Resnet34 and Imagenet for fair comparison.
  - Unet
  - DeepLabV3
  - Segformer

# 03

# Results


DeepLabV3
Segformer
Unet
Loss
IoU
F1
Precision
Recall
Train Loss
Val Loss
Val IoU
Val F1
Val Precision
Val Recall
Epoch

# Model Comparison



For fair comparison, I chose the 3 epochs with the lowest val_loss as a starting point. Then compare IoU and Dice(F1), with emphasis on F1 due to the sensitivity to smaller regions.

| | | epoch | train_loss | val_loss | val_iou | val_f1 | val_precision | val_recall |
|---|---|---|---|---|---|---|---|---|
| DeepLabV3 | 16 | 17 | 0.120685 | 0.125172 | 0.677902 | 0.728153 | 0.809013 | 0.808926 |
| | 17 | 18 | 0.133496 | 0.115188 | 0.639622 | 0.682570 | 0.736905 | 0.827456 |
| | 18 | 19 | 0.107787 | 0.129363 | 0.630527 | 0.667378 | 0.753778 | 0.754612 |
| Segformer | 14 | 15 | 0.100148 | 0.094091 | 0.795417 | 0.835350 | 0.901357 | 0.823175 |
| | 18 | 19 | 0.115937 | 0.108528 | 0.792785 | 0.835733 | 0.916811 | 0.816306 |
| | 19 | 20 | 0.093717 | 0.089745 | 0.782631 | 0.814096 | 0.928663 | 0.803607 |
| Unet | 14 | 15 | 0.136438 | 0.099671 | 0.670042 | 0.711164 | 0.751646 | 0.855558 |
| | 15 | 16 | 0.113434 | 0.106527 | 0.746843 | 0.789541 | 0.854570 | 0.828154 |
| | 16 | 17 | 0.106348 | 0.102671 | 0.751644 | 0.791835 | 0.856597 | 0.831246 |

# 04

# Evaluation

# Interpretation

To compare all models the following metrics are used:

- All val_loss above 1.0 is considered a greater loss.
- All val_loss below 0.19 is considered a lesser loss.

Best performing model: Segformer

| | epoch | train_loss | val_loss | val_iou | val_f1 | val_precision | val_recall |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 0.952247 | 1.064543 | 0.617380 | 0.631839 | 0.934760 | 0.650000 |
| 1 | 2 | 0.474234 | 3.720344 | 0.402323 | 0.472710 | 0.511285 | 0.559563 |
| 2 | 3 | 0.335997 | 1.366646 | 0.415014 | 0.483308 | 0.510932 | 0.627929 |
| 3 | 4 | 0.283524 | 0.492931 | 0.469693 | 0.504791 | 0.559813 | 0.646607 |
| 4 | 5 | 0.265486 | 0.301019 | 0.509283 | 0.561019 | 0.673049 | 0.700518 |
| 5 | 6 | 0.219889 | 0.230777 | 0.576130 | 0.611717 | 0.780927 | 0.702382 |
| 6 | 7 | 0.205397 | 1.027763 | 0.628357 | 0.649412 | 0.935356 | 0.660551 |
| 7 | 8 | 0.212334 | 0.811933 | 0.647358 | 0.668978 | 0.883356 | 0.681481 |
| 8 | 9 | 0.190104 | 0.315746 | 0.526947 | 0.603544 | 0.608241 | 0.810799 |
| 9 | 10 | 0.173847 | 0.251801 | 0.546765 | 0.612441 | 0.608201 | 0.780820 |
| 10 | 11 | 0.165668 | 0.262190 | 0.634392 | 0.659789 | 0.914693 | 0.666038 |
| 11 | 12 | 0.152444 | 0.193168 | 0.669258 | 0.706176 | 0.924055 | 0.701765 |
| 12 | 13 | 0.159632 | 0.131041 | 0.651669 | 0.701407 | 0.816326 | 0.803947 |
| 13 | 14 | 0.175177 | 0.141065 | 0.675401 | 0.717111 | 0.900434 | 0.753705 |
| 14 | 15 | 0.143512 | 0.188305 | 0.539041 | 0.579128 | 0.704830 | 0.718704 |
| 15 | 16 | 0.139891 | 0.153274 | 0.600402 | 0.651497 | 0.697219 | 0.809429 |
| 16 | 17 | 0.120685 | 0.125172 | 0.677902 | 0.728153 | 0.809013 | 0.808926 |
| 17 | 18 | 0.133496 | 0.115188 | 0.639622 | 0.682570 | 0.736905 | 0.827456 |
| 18 | 19 | 0.107787 | 0.129363 | 0.630527 | 0.667378 | 0.753778 | 0.754612 |
| 19 | 20 | 0.134435 | 0.202576 | 0.600730 | 0.659390 | 0.655943 | 0.859439 |

Segformer-

- Greater loss- 1 (3)
- Lesser loss- 10 (10-20)

| | epoch | train_loss | val_loss | val_iou | val_f1 | val_precision | val_recall |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 0.416282 | 0.340559 | 0.469205 | 0.487961 | 0.520972 | 0.652302 |
| 1 | 2 | 0.277832 | 0.811433 | 0.467503 | 0.482681 | 0.518273 | 0.649906 |
| 2 | 3 | 0.226012 | 0.823315 | 0.468977 | 0.511808 | 0.556128 | 0.709443 |
| 3 | 4 | 0.203121 | 2.667831 | 0.563397 | 0.583833 | 0.750037 | 0.644569 |
| 4 | 5 | 0.187067 | 0.509721 | 0.490120 | 0.552087 | 0.587737 | 0.684652 |
| 5 | 6 | 0.189770 | 0.393886 | 0.476439 | 0.535451 | 0.568338 | 0.702403 |
| 6 | 7 | 0.213712 | 0.944228 | 0.481078 | 0.522384 | 0.617877 | 0.669812 |
| 7 | 8 | 0.180256 | 0.366335 | 0.503271 | 0.552436 | 0.602927 | 0.705860 |
| 8 | 9 | 0.183581 | 0.208738 | 0.575271 | 0.633099 | 0.638673 | 0.778792 |
| 9 | 10 | 0.157013 | 0.235539 | 0.562316 | 0.630458 | 0.629615 | 0.834028 |
| 10 | 11 | 0.143330 | 0.334383 | 0.519752 | 0.599923 | 0.617008 | 0.781665 |
| 11 | 12 | 0.127008 | 0.145222 | 0.591691 | 0.650288 | 0.669422 | 0.797783 |
| 12 | 13 | 0.139535 | 0.124726 | 0.606323 | 0.653634 | 0.678558 | 0.783799 |
| 13 | 14 | 0.125719 | 0.113973 | 0.733585 | 0.777508 | 0.864345 | 0.808512 |
| 14 | 15 | 0.136438 | 0.099671 | 0.670042 | 0.711164 | 0.751646 | 0.855558 |
| 15 | 16 | 0.113434 | 0.106527 | 0.746843 | 0.789541 | 0.854570 | 0.828154 |
| 16 | 17 | 0.106348 | 0.102671 | 0.751644 | 0.791835 | 0.856597 | 0.831246 |
| 17 | 18 | 0.115446 | 0.118704 | 0.744745 | 0.785544 | 0.870310 | 0.814154 |
| 18 | 19 | 0.108643 | 0.112772 | 0.783658 | 0.823443 | 0.916763 | 0.808446 |
| 19 | 20 | 0.113623 | 0.114754 | 0.675489 | 0.720139 | 0.785457 | 0.819624 |

| | epoch | train_loss | val_loss | val_iou | val_f1 | val_precision | val_recall |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 1.069223 | 0.850776 | 0.453754 | 0.510438 | 0.543508 | 0.615170 |
| 1 | 2 | 0.399447 | 0.353780 | 0.483281 | 0.516094 | 0.596855 | 0.661233 |
| 2 | 3 | 0.249180 | 2.317963 | 0.279748 | 0.378481 | 0.550210 | 0.526056 |
| 3 | 4 | 0.206506 | 0.470016 | 0.599410 | 0.620767 | 0.743141 | 0.682236 |
| 4 | 5 | 0.161303 | 0.428346 | 0.579497 | 0.604933 | 0.860681 | 0.661884 |
| 5 | 6 | 0.219497 | 0.217069 | 0.550156 | 0.611835 | 0.631467 | 0.763530 |
| 6 | 7 | 0.166576 | 0.225718 | 0.553361 | 0.607425 | 0.652022 | 0.732391 |
| 7 | 8 | 0.164253 | 0.299598 | 0.633666 | 0.656286 | 0.940587 | 0.664534 |
| 8 | 9 | 0.160787 | 0.241642 | 0.500714 | 0.533588 | 0.760999 | 0.678910 |
| 9 | 10 | 0.139606 | 0.121405 | 0.619780 | 0.665040 | 0.733007 | 0.809761 |
| 10 | 11 | 0.108984 | 0.162445 | 0.665181 | 0.713418 | 0.774945 | 0.814236 |
| 11 | 12 | 0.108288 | 0.126845 | 0.738457 | 0.781924 | 0.855728 | 0.819206 |
| 12 | 13 | 0.112927 | 0.126398 | 0.628551 | 0.682064 | 0.721791 | 0.870191 |
| 13 | 14 | 0.096546 | 0.370081 | 0.537747 | 0.614387 | 0.625911 | 0.797669 |
| 14 | 15 | 0.100148 | 0.094091 | 0.795417 | 0.835350 | 0.901357 | 0.823175 |
| 15 | 16 | 0.117612 | 0.137846 | 0.629085 | 0.674409 | 0.664921 | 0.836043 |
| 16 | 17 | 0.113684 | 0.108853 | 0.608240 | 0.644089 | 0.700932 | 0.792709 |
| 17 | 18 | 0.115567 | 0.162287 | 0.699645 | 0.741695 | 0.933337 | 0.732708 |
| 18 | 19 | 0.115937 | 0.108528 | 0.792785 | 0.835733 | 0.916811 | 0.816306 |
| 19 | 20 | 0.093717 | 0.089745 | 0.782631 | 0.814096 | 0.928663 | 0.803607 |

DeepLabV3-

- Greater loss- 4 (1, 2,3, 6)
- Lesser loss- 8 (12-19)

Unet-

- Greater loss- 1 (4)
- Lesser loss- 9 (12-20)

# THANK YOU